

Software Requirements Specification

Fitup — Workout Planner, Training Log, and AI Training Assistant

Fitup Engineering

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1 Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the complete functional and non-functional requirements for **Fitup**, a local-first workout planner and training log for iOS, watchOS, and Android, together with the **AI Training Assistant** module and the supporting content-management and service infrastructure required to operate it.

The document specifies the system in full: the workout planning and training-log capabilities that form the product's core, and the AI Training Assistant that builds on them. It states what the system shall do and the conditions it shall satisfy. It does not prescribe how those requirements are to be realised, except where a constraint on implementation is itself a requirement — as is the case for the safety architecture in Section 5.13, where the arrangement of components is the control.

1.2 Intended audience

Audience	Sections of primary interest
Engineering	3, 4, 5, 6, 7, 9
Content operations	4.3, 5.14, 8
Product management	2, 5, 12, 13
Quality assurance	5, 10, Appendix B
Legal and compliance review	7.3, 8
Infrastructure and operations	6.3, 7.4, 9

1.3 Scope

1.3.1 In scope

Fitup consists of five deliverable components:

ID	Component	Status
C1	Mobile client (React Native / Expo), iOS and Android	Implemented
C2	watchOS companion application and iOS Live Activity surfaces	Implemented
C3	SyncLayer provider (optional server-side sync and catalogue distribution)	Contract defined; provider operated separately
C4	AI Training Assistant service	New
C5	Content Management System for exercise and nutrition catalogues	New

The product's defining architectural property is that it is **local-first**. The working database is SQLite on the device. Planning a workout, logging sets, and reviewing history never depend on a network connection or an account. All server-side components are optional enhancements layered over that local core, and the AI Assistant is the first feature in the product that is genuinely online-only.

1.3.2 Out of scope

The following are explicitly excluded from this specification:

- Conversational multi-turn chat with the assistant.
- Food photograph recognition and barcode scanning.
- Real-time form correction or coaching during a set.

- Social features: friends, feeds, sharing of plans between users.
- Any diagnostic, therapeutic, or medical claim (see Section 8).
- Web or desktop clients for end users. The CMS is web-based but is a staff tool.

1.4 Definitions, acronyms, and abbreviations

Term	Definition
Catalogue	The curated corpus of exercises and foods, authored in the CMS and distributed to clients.
Constraint	A hard rule a generated plan must satisfy, e.g. exclusion of spinal loading for a declared lumbar condition.
Generation	One end-to-end AI plan request. Consumes one unit of the user's monthly quota.
Local-first	Architecture in which the device database is authoritative for the user's own data and the network is optional.
Plan	An AI-generated proposal of workouts and, optionally, meals across a horizon of days.
Sync (feature)	The user action committing a generated plan into the workout schedule.
SyncLayer	The optional server-side data synchronisation provider.
System exercise	A catalogue exercise owned by Fitup, identified by the reserved user ID <code>__fitup__</code> .
Validator	The deterministic, non-AI component that verifies a generated plan against constraints.
BMR / TDEE	Basal Metabolic Rate / Total Daily Energy Expenditure.
RPE	Rate of Perceived Exertion, a subjective intensity scale recorded per set.
PR	Personal Record.

1.5 References

Ref	Document
R1	docs/ARCHITECTURE.md — client architecture
R2	docs/SYNC_PROTOCOL.md — sync provider contract
R3	docs/LOCAL_ONLY.md — building without a sync provider
R4	docs/RELEASING.md — release process
R5	src/db/schema/ — Drizzle schema definitions
R6	Apple App Review Guidelines, sections 1.4.1, 2.5.1, 5.1.1, 5.1.3
R7	Google Play Developer Program Policy — Health apps; Health Connect data-use policy
R8	Regulation (EU) 2016/679 (GDPR), Articles 6, 9, 15–22
R9	GNU General Public License, version 3.0

1.6 Document conventions

Requirement identifiers take the form `FR-<AREA>-<n>` for functional requirements and `NFR-<AREA>-<n>` for non-functional requirements. Each requirement is atomic and independently testable.

Priority is expressed with **Must** (required for release), **Should** (required for a complete product but may follow the initial release), and **May** (desirable).

The key words *must*, *must not*, *should*, and *may* are to be interpreted as described in RFC 2119.

2 Overall Description

2.1 Product perspective

Fitup is a self-contained mobile product rather than a component of a larger system. It depends on no external system for its core function, and it is this independence that the specification is principally concerned with preserving.

The system shall comprise:

- A tab-based mobile application with four primary sections — Home, Exercises, Results, and Settings — together with the detail, editor, and modal screens they lead to.
- An on-device relational database holding all user-owned records, versioned by migrations applied at application start.
- Native integrations for the watch companion, the live-session lock-screen surface, and the workout command channel between them.
- Health platform integration through HealthKit on iOS and Health Connect on Android.
- Localisation into English, Spanish, Hindi, Russian, and Chinese.
- Optional analytics, error reporting, and over-the-air update channels, each independently configurable and each disabled by default.
- The AI Training Assistant, together with the service and content-management components required to operate it.

The AI Assistant is distinguished from every other capability in this document by three properties: it is the only user-facing feature that requires a network connection, it is the only one that requires a structured profile of the user, and it is the only one whose output can cause physical harm if it is wrong. Each of these carries requirements that do not arise anywhere else in the product, and the specification treats the module accordingly rather than as an additive feature.

2.2 Product functions

At the highest level the system provides:

1. **Exercise catalogue management** — browse, search, filter, and fork a maintained catalogue of exercises; create and edit personal exercises.
2. **Workout planning** — compose workouts from exercises, organised into single exercises, supersets, trisets, and circuits, and schedule them.
3. **Live workout execution** — log sets with weight, repetitions, time, distance, and RPE; manage rest periods; control the session from an Apple Watch or the Dynamic Island.
4. **Body measurement tracking** — record measurements manually or import them from the platform health store; visualise trends.
5. **Results and analysis** — review history, training volume, personal records, and per-exercise statistics.
6. **AI plan generation** — produce a personalised training and nutrition plan from the user's profile, history, stated intent, and the curated catalogue, subject to hard safety constraints, and commit it to the schedule.
7. **Optional synchronisation** — replicate the local database to a compatible provider and receive the maintained exercise catalogue.
8. **Content operations** — author and govern the catalogue and its conditional safety metadata.

2.3 User classes and characteristics

Class	Description	Technical expertise	Frequency of use
Casual trainee	Follows a simple routine, logs sets, tracks weight. Primary audience.	Low	2–4 sessions/week
Experienced lifter	Uses supersets, RPE, drop sets; reviews volume and PRs closely.	Medium	4–6 sessions/week

Class	Description	Technical expertise	Frequency of use
Rehabilitation-constrained user	Trains around a declared condition. Highest safety sensitivity.	Low	Variable
Content editor	Client staff authoring catalogue entries and conditional rules.	Medium	Daily
Content reviewer	Senior staff approving safety-relevant rules before publication.	Medium; domain-qualified	Weekly
Support and operations	Fitup staff investigating generations, quota, and failures.	High	As required

The rehabilitation-constrained class deserves particular emphasis. It is the class the AI Assistant is most valuable to and the class it can most harm. Requirements throughout this document that concern contraindication handling exist to serve this class safely.

2.4 Operating environment

Attribute	Requirement
iOS	16.1 or later (Live Activities require 16.1; Dynamic Island requires supporting hardware)
watchOS	Paired Apple Watch, watchOS 9 or later
Android	API level 26 (Android 8.0) minimum; compile SDK 36, target SDK 35
Device storage	Approximately 150 MB application, plus catalogue media cached on demand
Network	Not required for core function; required for sync, catalogue updates, and AI generation
Backend	Linux containers on AWS; PostgreSQL 15 or later

2.5 Design and implementation constraints

ID	Constraint
DC-1	The core workout flow must remain fully functional with no network and no account. This constrains every feature added to the product, including the AI Assistant.
DC-2	The client is licensed GPL-3.0. Client modifications are derivative works and must remain GPL-3.0 and be published (see Section 8.4).
DC-3	The local database is SQLite via Drizzle ORM. Schema changes require generated migrations that run at application start.
DC-4	Sync shall use the documented contract with the <code>x-fitup-sync-schema</code> compatibility header. New server-owned data should reuse this contract rather than introduce a parallel mechanism.
DC-5	LLM provider credentials must never be present in the client binary, the repository, or the CMS.

ID	Constraint
DC-6	Health data obtained from HealthKit or Health Connect is subject to platform data-use restrictions that are stricter than the application's own privacy policy. The client is built and released through EAS; native module changes require a new binary and cannot be delivered over the air.
DC-7	

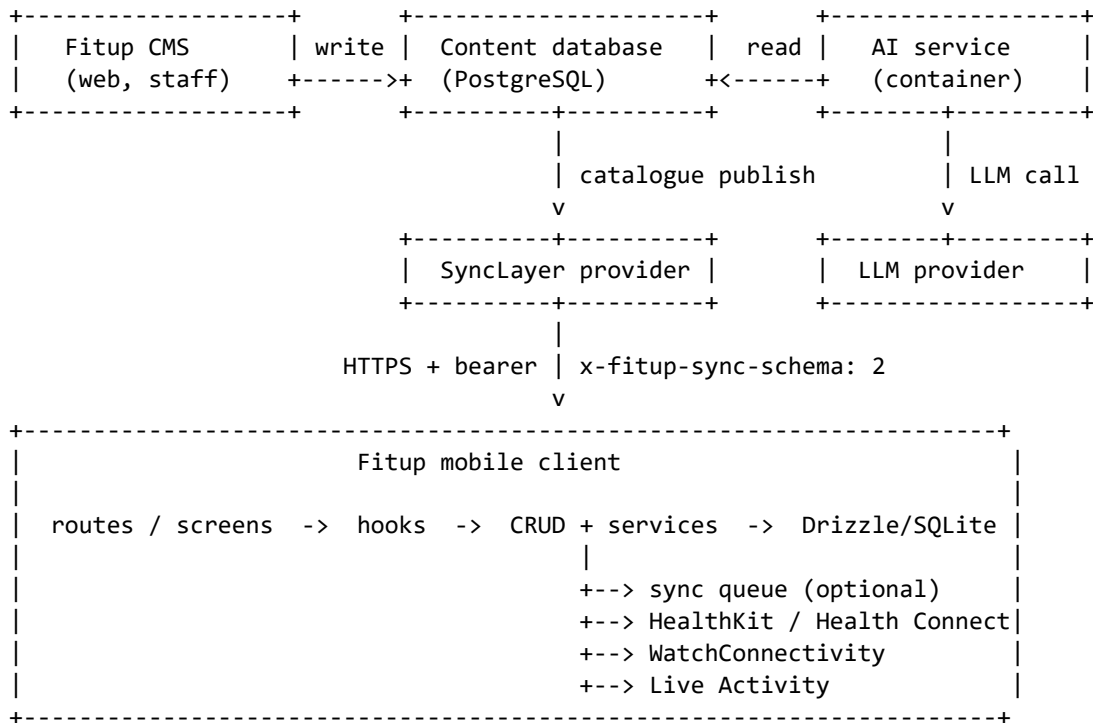
2.6 Assumptions and dependencies

ID	Assumption
A-1	Users have network connectivity at the moment of AI plan generation. Generation is the only online-only user-facing feature.
A-2	The catalogue is curated by qualified humans and is authoritative on safety. The model selects from it; it never invents an exercise or food.
A-3	Users self-declare conditions honestly, and may decline to declare any.
A-4	Generated plans are general fitness guidance, not medical guidance.
A-5	An LLM provider is available under commercial terms including zero data retention and no training on submitted data.
A-6	A qualified professional accepts named ownership of the clinical correctness of contraindication rules (see Section 13, Q4).

Assumption A-6 is currently **unsatisfied** and is the highest-severity open item in this document.

3 System Architecture

3.1 Architectural overview



3.2 Client layer model

The client follows a strict layering, described in R1 and preserved by this specification.

Layer	Location	Responsibility
Routes	src/routes/	Expo Router entry points; provider composition
Screens	src/screens/	Product screens, one directory per feature area
Components	src/components/	Reusable presentational and primitive UI
Hooks	src/hooks/	React state and data-access hooks
Queries	src/queries/	TanStack Query definitions
CRUD	src/crud/	All product writes; the single boundary at which sync enqueueing occurs
Services	src/services/	Platform integrations: health, watch, live activity, auth, notifications
Sync	src/sync/	Queue compaction, push, pull, backfill
DB	src/db/	Drizzle schema and connection
Stores	src/stores/	Zustand client state
Storage	src/storage/	MMKV key-value persistence

FR-ARC-1 — Must. All product writes must pass through src/crud/. Screens must not issue ad hoc SQL. This guarantees a single, consistent point at which sync-queue entries are created.

FR-ARC-2 — Must. The AI Assistant’s plan commit must use the CRUD functions for workout, workoutGroup, and workoutExercise. It must not introduce a second write path into the workout domain.

3.3 Technology stack

Concern	Technology
Application framework	React Native 0.86, React 19.2, Expo SDK 57
Navigation	Expo Router 57 (typed routes)
Local database	SQLite via expo-sqlite; Drizzle ORM 0.45
Server state	TanStack Query 5
Client state	Zustand 5
Key-value storage	react-native-mmkv 4
Styling	react-native-unistyles 3
Forms and validation	React Hook Form 7, Zod 4
Animation	Reanimated 4, Worklets
Charts	react-native-gifted-charts
Internationalisation	i18next 26, react-i18next 17, expo-localization
Health	@kingstinct/react-native-healthkit, react-native-health-connect
Analytics	PostHog (optional)
Diagnostics	Sentry (optional)
Backend service	Node.js or Python container; PostgreSQL 15
Model provider	OpenAI gpt-4o-mini (see 5.13.2)

4 Data Requirements

4.1 Core local schema

The device database shall hold the following tables. Primary keys shall be 21-character NanoIDs. All product tables shall carry `createdAt` and `updatedAt` timestamps in epoch milliseconds, and all schema changes shall be delivered as generated migrations applied at application start.

Table	Purpose	Notable columns
user	Device, locale, notification, and unit preferences	theme, weightUnits, distanceUnits, timeZone, notification capability flags
exercise	Exercise catalogue, both system and user-authored	category, tracking[], primaryMuscleGroups[], secondaryMuscleGroups[], equipment[], instructions[], mistakes[], muscleLoad, difficulty, gifFilename, source, fitupSourceId, userId
workout	A planned or executed session	status (planned/in_progress/completed/cancelled), startAt, startedAt, completedAt, duration, remind
workout_group	Grouping within a workout	type (single/superset/triset/circuit), order, notes
workout_exercise	An exercise instance within a workout	workoutId, exerciseId, groupId, orderInGroup
exercise_set	A single set	type (warmup/working/dropset/failure), weight, reps, time, distance, rpe, restTime, startedAt, completedAt
measurement	Body metrics, manual or imported	metric, value, unit, recordedAt, source (manual/health), sourcePlatform, externalId
app_review	Review-prompt state machine	—
sync_queue	Pending outbound operations	tableName, recordId, operation, data, timestamp, synced
sync_metadata	User-scope pull cursor	lastSyncTimestamp
fitup_sync_metadata	Catalogue pull cursor, keyed by locale	locale, lastSyncTimestamp

The exercise table carries the metadata surface on which intelligent selection depends — muscle groups, equipment, difficulty, and a structured `muscleLoad` profile. These attributes shall be populated for all catalogue content and shall be distributed to clients through the catalogue pull scope defined in 5.12.

4.2 Data domains required by the AI Assistant

The assistant depends on three data domains beyond the core schema. Each is a prerequisite for the module rather than a refinement of it, and together they represent a larger body of work than the generation logic itself. Section 12 schedules them accordingly.

4.2.1 Structured fitness profile

The core user table is scoped to device, locale, and unit preferences, and the measurement table to time-series body metrics. Neither is an appropriate home for goal, activity level, dietary pattern, equipment access, or declared condition, all of which are single-valued attributes of the person rather than device settings or dated observations.

A dedicated profile domain shall therefore be provided (4.3.1), together with the onboarding flow that populates it (5.2).

4.2.2 Nutrition

The assistant's nutrition output requires a food and meal domain: a catalogue of foods with macronutrient composition and dietary classification, and a structure for the meals composed from them. This domain shall support authoring in the CMS, distribution through the catalogue scope, generation, validation, and presentation (4.3.2).

4.2.3 Contraindication metadata

The exercise and food tables describe what an item *is* — muscles worked, equipment required, macronutrients supplied. Neither describes who an item is *unsafe for*.

A relation layer keyed to a controlled condition vocabulary shall therefore be provided (4.3.3). This layer is what the safety filtering in 5.13.4 operates on; without it, that filtering has nothing to filter against and the guarantee in 5.13.1.4 cannot be made.

4.3 Supporting schema

4.3.1 Profile

Table	Columns
user_profile	userId (PK), dateOfBirth, sex, heightCm, goal (lose/maintain/gain/recomp), activityLevel, sessionsPerWeek, sessionMinutes, dietaryPattern, derivedBmr, derivedTdee, profileVersion, timestamps
user_equipment	userId, equipmentKey — the equipment the user can actually access
user_condition	userId, conditionKey, declaredAt — self-declared conditions from the controlled vocabulary
user_allergen	userId, allergenKey

4.3.2 Nutrition

Table	Columns
food	id (PK), name, servingSize, servingUnit, kcal, proteinG, carbsG, fatG, fibreG, tags[], dietaryFlags[], allergens[], source, userId, timestamps
meal	id (PK), userId, date, slot (breakfast/lunch/dinner/snack), order, timestamps
meal_item	id (PK), mealId, foodId, servings, timestamps

4.3.3 Catalogue governance

Table	Columns
condition	key (PK), label, description, severity — the controlled vocabulary
exercise_condition	exerciseId, conditionKey, relation (indicated/contraindicated), note
food_condition	foodId, conditionKey, relation, note

4.3.4 AI

Table	Columns
ai_plan	id (PK), userId, intent, note, horizonDays, planJson, modelId, promptVersion, validatorResult, createdAt, committedAt, expiresAt
ai_quota_ledger	id (PK), userId, periodStart, consumed, limit, updatedAt — server-side only

FR-DAT-1 — Must. ai_quota_ledger must exist only on the server. It must never be replicated to the device, because a client-side quota record is a client-side quota control (see 5.13.7).

FR-DAT-2 — Must. condition, exercise_condition, and food_condition are server-owned catalogue content. They must be distributed through the existing fitup pull scope, not authored on the device.

FR-DAT-3 — Must. All new user-owned tables must participate in the existing sync queue using the established src/crud/ boundary.

5 Functional Requirements

5.1 Identity and accounts

FR-IDN-1 — Must. The application must be fully usable without an account. A local user record is created on first launch.

FR-IDN-2 — Must. When a sync provider is configured, the client obtains a bearer token via `POST /auth/token` using the local user ID and a persistent device ID, and stores it in MMKV with an in-memory fallback for the current launch.

FR-IDN-3 — Must. On a 401 response the client must clear the stored token, request a new one, and retry the failed request exactly once.

FR-IDN-4 — Must. AI generation requires an authenticated user. If no provider is configured, the AI entry point must be hidden rather than shown in a failing state.

5.2 Onboarding and profile

FR-PRO-1 — Must. Onboarding must capture date of birth, sex, height, current weight, target weight, goal, activity level, sessions per week, and typical session length.

FR-PRO-2 — Must. Onboarding must capture available equipment from the controlled equipment vocabulary defined for `exercise.equipment`.

FR-PRO-3 — Must. Onboarding must capture dietary pattern and allergen exclusions.

FR-PRO-4 — Must. Onboarding must capture self-declared conditions from the controlled vocabulary, with an explicit and clearly presented *none* option.

FR-PRO-5 — Must. Condition and allergen capture must be optional. A user who declines must still be able to generate a plan; the plan is then generated under general-population constraints.

FR-PRO-6 — Must. Onboarding must be skippable in full. A user who skips retains complete access to all non-AI functionality, in accordance with DC-1.

FR-PRO-7 — Must. Every profile field must be editable afterwards from Settings. Editing must increment `profileVersion` and invalidate any cached generation context.

FR-PRO-8 — Must. Height and weight must continue to write to `measurement` so that existing charts, health import, and history remain correct and unduplicated.

FR-PRO-9 — Should. The client should derive BMR and TDEE locally using the Mifflin-St Jeor equation and transmit the derived values rather than the raw inputs where the raw inputs are not otherwise required.

5.3 Exercise catalogue

FR-EXC-1 — Must. Users must be able to browse, search, and filter exercises by category, muscle group, and equipment.

FR-EXC-2 — Must. Users must be able to create, edit, and delete personal exercises.

FR-EXC-3 — Must. System exercises (`userId = '__fitup__'`) must not be directly editable or deletable. Editing a system exercise must fork it into a user-owned copy with `fitupSourceId` set to the source, and existing workout references must be remapped to the fork.

FR-EXC-4 — Must. Exercise animations must load from the configured media base URL at thumbnail (180 px) and preview (1080 px) resolutions. The base URL must be overridable by environment variable.

FR-EXC-5 — Should. Exercise detail must present instructions, common mistakes, primary and secondary muscle groups, equipment, and difficulty.

FR-EXC-6 — Must. Exercise detail must display declared contraindications when the viewing user has a matching declared condition, with a clear, non-alarming explanation.

5.4 Workout planning

FR-WKP-1 — Must. Users must be able to create a workout, name it, add exercises, and reorder them.

FR-WKP-2 — Must. Exercises must be groupable as single, superset, triset, or circuit, with ordering preserved within the group.

FR-WKP-3 — Must. Workouts must be schedulable via `startAt`, with an optional reminder offset of start, 5 m, 10 m, 15 m, 30 m, 1 h, or 2 h.

FR-WKP-4 — Must. A workout must hold exactly one of the statuses `planned`, `in_progress`, `completed`, or `cancelled`.

FR-WKP-5 — Should. Users must be able to duplicate an existing workout.

5.5 Live workout execution

FR-WKE-1 — Must. Starting a workout must transition it to `in_progress` and record `startedAt`.

FR-WKE-2 — Must. Sets must support the types `warmup`, `working`, `dropset`, and `failure`.

FR-WKE-3 — Must. Set logging must support weight, repetitions, time, distance, and RPE according to the parent exercise's tracking configuration.

FR-WKE-4 — Must. Completing a set must record `completedAt` and, where a rest time is configured, begin the rest timer.

FR-WKE-5 — Must. Rest timers must support configurable duration, sound, and haptic feedback, and must survive application backgrounding.

FR-WKE-6 — Must. All writes during a live workout must commit to SQLite before any network activity is attempted.

FR-WKE-7 — Should. The screen must optionally remain awake during a workout, under user control.

FR-WKE-8 — Must. Completing a workout must set `completedAt` and compute duration.

5.6 Body measurements

FR-MEA-1 — Must. Users must be able to record body metrics manually with a value, unit, and timestamp.

FR-MEA-2 — Must. Authorised measurements must import from HealthKit or Health Connect, recorded with `source = 'health'` and the originating platform.

FR-MEA-3 — Must. Imported measurements must be deduplicated on (`userId`, `source`, `metric`, `externalId`).

FR-MEA-4 — Should. Measurements must be presented as trend charts over a selectable range.

5.7 Results and analysis

FR-RES-1 — Must. Users must be able to review completed workouts by day and by month.

FR-RES-2 — Must. The system must compute per-exercise history, training volume, and personal records.

FR-RES-3 — Should. Results must present activity summaries, strength progression, and body-weight trend.

FR-RES-4 — Should. Where a workout originated from an AI plan, results must indicate this and link to the originating plan.

5.8 Health platform integration

FR-HLT-1 — Must. On iOS the client must read authorised HealthKit data and may write completed workouts.

FR-HLT-2 — Must. On Android the client must integrate Health Connect equivalently.

FR-HLT-3 — Must. Absent services or denied permissions must never break the local workout flow. All health integration is strictly additive.

FR-HLT-4 — Should. Heart-rate zones must be computed from a user-selected maximum-heart-rate formula: Nes, Fox, Tanaka, Inbar, Gulati, Gellish, or a manual value.

FR-HLT-5 — Must. Health-derived data must not be transmitted to the LLM provider unless the user has given separate, explicit, informed consent. See FR-CMP-6.

5.9 Apple Watch and Live Activities

FR-WAT-1 — Must. The watchOS application must display the active workout and permit set completion and rest control.

FR-WAT-2 — Must. Watch and phone state must remain consistent through WatchConnectivity messaging.

FR-WAT-3 — Must. On iOS 16.1 and later the active workout must present a Live Activity, including Dynamic Island on supporting hardware.

FR-WAT-4 — Should. The Live Activity must expose set completion and rest control as interactive intents.

5.10 Notifications

FR-NOT-1 — Must. Scheduled workouts must raise a local notification at the configured reminder offset.

FR-NOT-2 — Must. Rest completion must raise a notification when the application is backgrounded.

FR-NOT-3 — Must. All notification categories must be individually controllable in Settings.

5.11 Settings, units, and localisation

FR-SET-1 — Must. Theme must be selectable as light, dark, or system.

FR-SET-2 — Must. Units must be independently configurable: body weight (kg/lb), training weight (kg/lb), distance (km/mi), measurement (cm/in), temperature (C/F).

FR-SET-3 — Must. The interface must be localised into English, Spanish, Hindi, Russian, and Chinese, following the device locale by default with manual override.

FR-SET-4 — Must. Date, time, first weekday, and time format must be configurable.

FR-SET-5 — Must. Settings must expose AI Assistant controls: remaining quota, generation history, profile editing, and a complete opt-out.

FR-SET-6 — Must. Opting out of the AI Assistant must delete the stored profile and generation history and hide the AI entry point entirely.

5.12 Synchronisation

FR-SYN-1 — Must. Sync must be entirely optional, selected at build time by EXPO_PUBLIC_SYNC_HOST. Without it the client must not mount the sync provider, authenticate, or enqueue operations.

FR-SYN-2 — Must. Local create, update, and delete operations must enqueue to sync_queue; the engine must compact compatible operations per record before transmission.

FR-SYN-3 — Must. Requests must carry x-fitup-sync-schema: 2. A provider must reject an unsupported schema version rather than silently accept an incompatible payload.

FR-SYN-4 — Must. The maintained catalogue must pull through a separate fitup scope with a per-locale cursor, distinct from the user-data cursor.

FR-SYN-5 — Must. User-authored exercises must remain distinguishable from catalogue exercises by source and userId.

FR-SYN-6 — Must. New nutrition and profile tables must extend the existing protocol under the same schema-version discipline. A schema change requires incrementing the header value.

5.13 AI Training Assistant

This section specifies the module that is the subject of the current development phase. It begins with four findings that alter the design proposed in the originating brief.

5.13.1 Findings that alter the proposed design

5.13.1.1 Model parameters as briefed are invalid The brief specifies “*temperature 200 and 0.4 as token limit.*” These values are transposed, and the token figure is unusable in either arrangement.

Parameter	As briefed	Valid range	Specified
temperature	200	0.0–2.0	0.4
max_output_tokens	0.4	1–16,384	3,000

A temperature of 200 is rejected by the API. More consequentially, **200 output tokens cannot contain a plan**. Two hundred tokens is roughly 150 words — less than one meal entry with macronutrients. A seven-day plan with exercises, sets, meals, and rationale measures 1,800–2,800 output tokens in practice.

Temperature 0.4 is a sound choice and is adopted: low enough for consistent structured output, high enough that a user does not receive an identical plan every month.

5.13.1.2 A client-only design cannot ship The brief proposes handling generation in the client, with “*not a lot of backend APIs.*” This is not viable for three independent reasons.

1. **Credential exposure.** Any secret bundled into a React Native binary — environment variable, obfuscated constant, or native module — is extractable from the IPA or APK. A leaked key is billed to the account holder until revoked and is discovered by automated scanners within hours of publication.
2. **Unenforceable quota.** A monthly limit enforced in client code is enforced by the attacker. Quota must be counted server-side against an authenticated identity.
3. **Catalogue distribution.** See 5.13.1.3.

A server component is therefore mandatory. It is, however, genuinely small: one authenticated endpoint performing a quota check, a retrieval query, a model call, and a validation pass. This is not the broad API surface the brief sought to avoid, and the client remains local-first for every other feature.

5.13.1.3 The catalogue cannot be placed in the system prompt The brief proposes including “*a very big catalog of foods and everything*” in the system prompt. At a realistic 5,000 foods at approximately 30 tokens each, this is roughly 150,000 tokens — **exceeding the 128,000-token context window of gpt-4o-mini**, so the request fails outright rather than degrading.

Even where it fits, the economics are decisive:

Approach	Input tokens per generation	Cost per generation	10,000 users x 3/month
Full catalogue in prompt	~150,000	~\$0.024	~\$675/month
Retrieval (specified)	~4,000	~\$0.0024	~\$72/month

Retrieval is specified in 5.13.4. It also improves quality: a smaller, better-targeted candidate set yields more consistent selection than a corpus the model must scan.

5.13.1.4 Constraint adherence cannot be delegated to the model The brief states that “*AI must always follow the constraints.*” No language model offers that guarantee. Prompting reduces violation rates; it does not eliminate them. Because this module recommends physical exercise to users who have declared conditions such as lumbar injury, a violation is a physical-safety event and a store-compliance event.

A deterministic validator is therefore specified in 5.13.5. **The model proposes; the validator disposes.** No plan reaches a user in a violating state.

5.13.2 Model configuration

Setting	Value
Model	gpt-4o-mini
temperature	0.4
max_output_tokens	3,000
response_format	Strict JSON Schema
Timeout	30 s
Retry	One, on transport error or schema-validation failure only

FR-AIA-1 — Must. The model identifier must be a runtime configuration value, not a compile-time constant.

FR-AIA-2 — Should. At least one alternative model must be benchmarked against the Section 10 acceptance suite before launch. Provider selection should follow the measured constraint-violation rate, which is the metric that governs safety here, rather than list price or general benchmark performance.

5.13.3 Prompt composition

Four layers, assembled server-side:

1. **System layer** — role definition, hard rules, refusal conditions, and output contract. Static and versioned.
2. **Few-shot layer** — three to five curated exemplars covering a straightforward bulking plan, a cut with dietary exclusions, and a plan with an active contraindication. Static and versioned.
3. **Retrieved context** — the candidate exercises and foods for this specific request.
4. **User context** — profile, derived energy targets, recent training history, and the stated intent.

FR-AIA-3 — Must. Layers 1 and 2 must be versioned artefacts held in the repository and changed only through code review. A prompt change is a release and carries the same process weight as a code change.

FR-AIA-4 — Must. Every generation record must store the prompt version used, so that any plan can be reproduced and any regression bisected.

5.13.4 Retrieval

FR-AIA-5 — Must. The service must resolve the user's hard exclusions — declared conditions, allergens, dietary pattern, and absent equipment — into a database filter.

FR-AIA-6 — Must. Candidate selection must apply that filter **in SQL**. Excluded items must never be included in the prompt. *This is the primary safety mechanism; the prompt is secondary.*

FR-AIA-7 — Must. Only published catalogue content may be retrieved.

FR-AIA-8 — Should. Remaining candidates should be ranked by goal fit and by variety relative to recent training history, taking approximately the top 60 exercises and 120 foods, tuned to hold assembled input near 4,000 tokens.

The governing property is that **an item a user must not receive is removed before the model is aware of it**. A model cannot select what it was never shown.

5.13.5 Validation

FR-AIA-9 — Must. Every generated plan must pass a deterministic validator before display. The validator is ordinary code and must not invoke a model.

Check	Action on failure
Every exerciseId and foodId exists in the supplied candidate set	Reject
No item contraindicated for a declared condition	Reject
No item violating a declared allergen or dietary pattern	Reject
No equipment the user does not have	Repair — substitute nearest permitted item
Daily energy within 10% of derived target	Repair — adjust servings
Protein within 15% of target	Repair
Weekly session count matches stated availability	Repair — drop lowest-priority day
No consecutive-day loading of the same primary muscle group	Repair — reorder

FR-AIA-10 — Must. A rejection must trigger exactly one regeneration with the violation fed back into context. A second rejection must fail the request.

FR-AIA-11 — Must. A failed request must not consume quota.

FR-AIA-12 — Must. Every rejection must be logged with the violating item and the rule violated, so that content operations can locate the catalogue gap responsible.

5.13.6 Generation flow

FR-AIA-13 — Must. The user initiates generation from a visible entry point and states an intent.

FR-AIA-14 — Must. Intent must be selected from presets, with an optional free-text note limited to 200 characters.

FR-AIA-15 — Must. Quota must be checked before the model is called.

FR-AIA-16 — Must. The plan must be presented as a preview, not committed, with a per-item rationale explaining why each item was selected.

FR-AIA-17 — Must. The preview must display remaining quota and the reset date.

FR-AIA-18 — Should. Generation must be cancellable, and a cancelled generation must not consume quota.

FR-AIA-19 — Should. The most recent plan must be retained for 30 days and be re-viewable at no quota cost.

5.13.7 Quota and cost control

FR-AIA-20 — Must. Default quota is three successful generations per calendar month per user.

FR-AIA-21 — Must. Quota must be enforced server-side against the authenticated user.

FR-AIA-22 — Must. Quota must be configurable per user and per cohort without an application release.

FR-AIA-23 — Must. Exhausted quota must return a distinct, non-error state carrying the reset date, and must be presented as an expected condition rather than a failure.

FR-AIA-24 — Must. A global daily spend ceiling must halt generation and alert operations.

FR-AIA-25 — Must. A per-user rate limit of one in-flight generation and five per hour must apply independently of the monthly quota.

5.13.8 Commit to schedule

FR-AIA-26 — Must. The *Sync* action must write the plan into *workout*, *workout_group*, and *workout_exercise* within a single database transaction.

FR-AIA-27 — Must. The user must choose **replace** or **append** for the target date range. The system must never silently overwrite an existing schedule.

FR-AIA-28 — Must. Replace must affect only workouts with `status = 'planned'`. Workouts that are `in_progress`, `completed`, or `cancelled` must never be modified. This requirement protects the user's training history, which is the most valuable data in the application and is irreplaceable.

FR-AIA-29 — Must. Committed rows must enqueue through the sync queue. No new sync path may be introduced.

FR-AIA-30 — Must. The commit must be undoable for 24 hours, restoring the previous schedule exactly.

FR-AIA-31 — Must. Plan exercises must resolve to existing catalogue rows. The client must never create an exercise from model output.

FR-AIA-32 — Must. Where a referenced exercise is absent locally, the client must fetch it before committing or fail the commit atomically. A partially committed plan is not an acceptable outcome.

FR-AIA-33 — Should. Committed workouts must record their originating `ai_plan.id` for traceability.

5.14 Content Management System

FR-CMS-1 — Must. Editors must be able to create, edit, archive, and search exercises and foods.

FR-CMS-2 — Must. Each food must record name, serving size and unit, energy, protein, carbohydrate, fat, and fibre.

FR-CMS-3 — Must. Each food must record dietary flags covering at minimum vegetarian, vegan, halal, gluten-free, dairy-free, and nut-free.

FR-CMS-4 — Must. Each exercise must record `indicatedFor` and `contraindicatedFor` relations against the controlled condition vocabulary.

FR-CMS-5 — Must. The condition vocabulary must itself be CMS-managed. Editors must not be able to enter free-text conditions, because free text cannot be filtered on reliably and would silently defeat FR-AIA-6.

FR-CMS-6 — Must. Content must follow a `draft -> in_review -> published` lifecycle. Only published content may reach generation.

FR-CMS-7 — Must. Publication of safety-relevant metadata must require a second, distinct approver.

FR-CMS-8 — Must. All changes must be audit-logged with actor, timestamp, and a before/after difference.

FR-CMS-9 — Should. Editors should be able to preview the population a rule would affect before publishing it.

FR-CMS-10 — Should. Bulk CSV import must be supported, with per-row validation and a dry-run mode.

FR-CMS-11 — Must. The CMS must be accessible only to authenticated staff, with roles distinguishing editor, reviewer, and administrator.

6 External Interface Requirements

6.1 User interfaces

FR-UI-1 — Must. Navigation is tab-based: Home, Exercises, Results, and Settings, with modal and stack presentation for detail and editor screens.

FR-UI-2 — Must. The interface must support light and dark themes and respect the system setting when configured to do so.

FR-UI-3 — Must. The interface must render correctly edge-to-edge and respect safe-area insets, including devices with a Dynamic Island.

FR-UI-4 — Must. The AI entry point must be discoverable from Home without displacing the primary workout action.

FR-UI-5 — Must. Plan preview must clearly distinguish proposed content from committed content, so that a user is never uncertain whether their schedule has changed.

6.2 Hardware interfaces

Interface	Requirement
Apple Watch	WatchConnectivity session for bidirectional workout control
Haptic engine	Rest and set feedback, subject to the user's haptics setting
Audio	Rest-completion tones, subject to the volume setting and silent-switch behaviour

6.3 Software interfaces

6.3.1 Sync provider

Defined normatively in R2. Summarised here for completeness.

Endpoint	Purpose
POST /auth/token	Exchange local user ID and device ID for a bearer token and expiry
POST /sync	Push pending local changes, grouped by table name into created, updated, deleted
GET /sync	Pull changes after the stored cursor; scope all, user, or fitup

All requests carry `x-fitup-sync-schema: 2` and `Authorization: Bearer <token>`.

6.3.2 AI service

POST /v1/plan/generate — authenticated, bearer token.

```
{
  "intent": "bulk",
  "note": "training around lower back pain",
  "profileVersion": "sha256:...",
  "horizonDays": 7
}
```

Status	Meaning	Body
200	Plan generated and validated	planId, plan, quotaRemaining, quotaResetsAt
429	Quota exhausted — an expected state, not an error	quotaRemaining, quotaResetsAt
422	No valid plan could be produced	detail, quotaConsumed: false
503	Model provider unavailable	retryAfter

Supporting endpoints:

Endpoint	Purpose
GET /v1/plan/quota	Quota state without generating
GET /v1/plan/{planId}	Re-fetch a retained plan
GET /v1/catalogue/sync	Incremental catalogue pull

6.3.3 Plan output contract

```
{
  "planId": "string",
  "horizonDays": 7,
  "rationale": "string",
  "days": [{
    "dayIndex": 0,
    "workout": {
      "name": "string",
      "exercises": [{
        "exerciseId": "string",
        "sets": 3,
        "reps": 10,
        "restSeconds": 90,
        "rationale": "string"
      }]
    },
    "meals": [{
      "slot": "breakfast",
      "items": [{ "foodId": "string", "servings": 1.5 }]
    }],
    "targets": { "kcal": 2600, "proteinG": 180, "carbsG": 300, "fatG": 70 }
  }]
}
```

FR-INT-1 — Must. exerciseId and foodId are references into the supplied candidate set. The model must not emit names, macronutrient values, or instructions; these are joined from the database at render time.

This constraint eliminates an entire class of failure. The model cannot hallucinate a food’s energy content, because it never writes one.

6.4 Communication interfaces

NFR-COM-1 — Must. All network communication must use TLS 1.2 or later.

NFR-COM-2 — Must. Certificate validation must not be disabled in any build configuration, including development builds.

NFR-COM-3 — Must. Requests must time out and fail gracefully without blocking the user interface.

7 Non-Functional Requirements

7.1 Performance

ID	Requirement
NFR-PER-1	Cold start to interactive must not exceed 2.5 s on a mid-tier device.
NFR-PER-2	Set logging must commit and reflect in the interface within 100 ms.
NFR-PER-3	Exercise list scrolling must sustain 60 fps at 1,000 catalogue entries.
NFR-PER-4	AI generation must complete within 12 s at the 95th percentile.
NFR-PER-5	Plan commit must complete within 500 ms for a seven-day plan.
NFR-PER-6	Migrations must complete within 3 s on the largest supported database.

7.2 Security

ID	Requirement
NFR-SEC-1	LLM provider credentials must exist only in the AI service environment — never in the client, repository, or CMS.
NFR-SEC-2	Bearer tokens must be stored in MMKV and must never be written to logs or analytics.
NFR-SEC-3	The AI service must be stateless; all state resides in PostgreSQL or the quota ledger.
NFR-SEC-4	All AI service inputs must be validated against a schema before use.
NFR-SEC-5	The free-text intent note must be treated as untrusted input and must not be able to alter system-prompt behaviour.
NFR-SEC-6	CMS access must require authentication with role-based authorisation and must be audit-logged.
NFR-SEC-7	Secrets must be held in a managed secret store and injected at runtime, never baked into container images.

NFR-SEC-5 deserves emphasis. The free-text note is user-controlled text placed into a model prompt — a prompt-injection surface. A user could attempt to instruct the model to disregard its constraints. The architecture contains this risk: because contraindicated items are filtered in SQL before the model is invoked (FR-AIA-6) and the validator re-checks the output (FR-AIA-9), a successful injection cannot cause an excluded item to be recommended. This is the practical benefit of not relying on the prompt for safety.

7.3 Privacy and data protection

ID	Requirement
NFR-PRV-1	Health and condition data must be encrypted at rest.
NFR-PRV-2	Condition, allergen, and health data must never be included in analytics events.

ID	Requirement
NFR-PRV-3	Data transmitted to the model provider must be governed by an explicit allow-list of fields, never a deny-list.
NFR-PRV-4	Users must be able to delete their profile and generation history; deletion must propagate within 30 days.
NFR-PRV-5	Users must be able to export their data in a machine-readable format.
NFR-PRV-6	Analytics must remain disabled unless explicitly configured, and must be disabled by default in development and preview builds.
NFR-PRV-7	The model provider must be contractually bound to zero retention and no training on submitted data.

NFR-PRV-3 is a deliberate design choice. An allow-list fails closed: a newly added profile field is excluded from transmission until someone consciously adds it. A deny-list fails open, silently transmitting every field added thereafter.

7.4 Reliability and availability

ID	Requirement
NFR-REL-1	Core workout functionality must be fully available offline.
NFR-REL-2	AI service unavailability must not degrade any other feature.
NFR-REL-3	Sync failures must be retried with exponential backoff and must never lose queued operations.
NFR-REL-4	Interrupted migrations must not corrupt the database; a failed migration must leave the previous schema intact.
NFR-REL-5	The AI service must target 99.5% monthly availability.
NFR-REL-6	Model provider outage must surface as a distinct, retryable state and must not consume quota.

7.5 Usability and accessibility

ID	Requirement
NFR-USA-1	Interactive targets must be at least 44 x 44 points.
NFR-USA-2	Text must respect the platform dynamic-type setting.
NFR-USA-3	Text contrast must meet WCAG 2.1 AA in both themes.
NFR-USA-4	All interactive elements must carry accessibility labels.
NFR-USA-5	Destructive actions must require confirmation and must be reversible where technically feasible.
NFR-USA-6	AI-generated content must be visually distinguishable from user-authored content.

7.6 Maintainability

ID	Requirement
NFR-MNT-1	<code>bun run verify</code> — lint, typecheck, and tests — must pass on every commit.
NFR-MNT-2	Product writes must remain confined to <code>src/crud/</code> .
NFR-MNT-3	Schema changes must ship as generated Drizzle migrations; hand-edited migrations are prohibited.
NFR-MNT-4	Prompt and few-shot artefacts must be versioned and independently rollback-able from application releases.
NFR-MNT-5	Text files must use LF endings, enforced by <code>.gitattributes</code> , because the formatter is configured for LF.

7.7 Observability

ID	Requirement
NFR-OBS-1	Every generation must be logged with input hash, model identifier, prompt version, token counts, cost, validator outcome, and latency.
NFR-OBS-2	Validator rejections must be logged at item and rule granularity.
NFR-OBS-3	Cost per generation and aggregate daily spend must be dashboarded.
NFR-OBS-4	Alerts must fire on daily spend ceiling, validator rejection rate, 95th-percentile latency, and error rate.
NFR-OBS-5	Logs must never contain personal health data.

8 Compliance, Legal, and Store Policy

The AI Assistant materially changes Fitup’s regulatory and store-review posture. These requirements are launch blockers, not follow-up work.

8.1 Medical positioning

ID	Requirement
FR-CMP-1	A persistent disclaimer must state that plans are general fitness guidance and not medical advice. It must be presented before first generation and must not be dismissible on first view.
FR-CMP-2	The application must make no diagnostic, treatment, or curative claim. Condition data is used solely to <i>exclude</i> content, never to <i>address</i> a condition.
FR-CMP-3	Users with a declared condition must be advised to consult a qualified professional before beginning.
FR-CMP-4	Interface and store copy must avoid “treats”, “heals”, “fixes”, “cures”, “therapy”, and “rehabilitation”.

The distinction in FR-CMP-2 is the one on which the product’s regulatory position rests. A system that *excludes* a squat from a plan because the user declared a lumbar condition is applying a safety filter. A system that *prescribes* an exercise to *treat* that condition is making a medical claim and would attract regulatory obligations the product is not built to satisfy. Every piece of copy must respect this line.

8.2 Store policy

ID	Requirement
FR-CMP-5	The privacy policy must disclose transmission of profile data to a named third-party model provider, and state the retention period.
FR-CMP-6	HealthKit and Health Connect data must not be transmitted to the model provider without separate explicit consent. Apple guideline 5.1.1(i) prohibits use of health data for advertising or sale; Health Connect restricts onward transmission.
FR-CMP-7	Use of AI must be disclosed on both store listings.
FR-CMP-8	Provider terms including zero retention and no training must be contracted before launch.
FR-CMP-9	User-generated free-text must not be publishable to other users, avoiding user-generated-content moderation obligations.

8.3 Data protection

ID	Requirement
FR-CMP-10	Health and condition data constitutes special-category data under GDPR Article 9. Processing requires explicit consent, obtained separately from general terms acceptance.

ID	Requirement
FR-CMP-11	Users must be able to exercise access, rectification, erasure, and portability rights in-application.
FR-CMP-12	A record of processing activities must be maintained covering the model provider as a processor.
FR-CMP-13	A Data Protection Impact Assessment must be completed before EU launch, since the feature involves automated processing of special-category data.

8.4 Licensing

The Fitup client is licensed under GPL-3.0 (R9), and is a rebranded derivative of the Skulpt workout tracker distributed under the same licence.

ID	Requirement
FR-CMP-14	Client modifications for this module are derivative works and must remain GPL-3.0 and be published.
FR-CMP-15	The full licence text and attribution must be retained in the repository and reproduced in-application.
FR-CMP-16	The separability of the CMS and AI service as independent works communicating over a network must be confirmed with counsel before any part of the stack is treated as proprietary.

FR-CMP-16 is not a formality. The conventional reading is that a separate program communicating with a GPL program over a network interface is not thereby derivative, but the analysis is fact-specific and turns on the degree of integration. It should be settled before commercial commitments depend on the answer.

9 Infrastructure and Deployment

9.1 Load characteristics

Sizing follows from actual expected load rather than from headroom estimates.

At 10,000 monthly active users generating three plans each, the system serves approximately **30,000 generations per month**, or roughly 1,000 per day. Concentrated into peak evening hours, this is approximately **two to five concurrent requests**. Each request spends nearly all of its wall-clock time awaiting the model provider, not computing.

9.2 Platform options

Option	Approximate monthly cost	Assessment
ECS Fargate, 2 x 0.5 vCPU	~\$25	Recommended. No hosts to patch; scales down cleanly.
Single EC2 t4g.small with Compose	~\$12	Acceptable and cheapest. Patching and restarts become your responsibility.
EKS	~\$73 control plane, plus nodes; ~\$150+	Not recommended at this scale.

The brief raises EC2 with Docker, or EKS. At five concurrent requests, **the EKS control plane alone would cost more than the model usage it orchestrates**. Kubernetes earns its operational overhead when many services are run by a team with established practice; it should be adopted at that point, not for a single stateless endpoint.

PostgreSQL should be RDS rather than self-managed. Automated backups and point-in-time recovery are worth more than the saving, particularly for the catalogue, which represents substantial irreplaceable human curation effort.

9.3 Deployment requirements

ID	Requirement
NFR-DEP-1	The service must be containerised, with images built in CI and tagged by commit.
NFR-DEP-2	Secrets must be held in AWS Secrets Manager and injected at runtime.
NFR-DEP-3	Deployments must be rolling or blue/green, with prompt versions pinned per deployment.
NFR-DEP-4	Structured JSON logs must ship to CloudWatch with cost and validator metrics.
NFR-DEP-5	Database migrations must run as a separate, explicitly triggered step, not on service start.
NFR-DEP-6	The catalogue must be backed up daily with a tested restore procedure.

9.4 Client release

ID	Requirement
NFR-DEP-7	Client builds ship through EAS with development, preview, and production profiles.
NFR-DEP-8	JavaScript-only changes may ship as over-the-air updates; native module changes require a store submission.

ID	Requirement
NFR-DEP-9	Source maps must upload to Sentry on every production build.
NFR-DEP-10	Runtime version policy follows <code>appVersion</code> , so an OTA update can never target an incompatible native binary.

10 Verification and Acceptance

10.1 Verification approach

Automated test coverage shall be maintained for sync flow, catalogue forking, analytics, authentication, health statistics, notification chains, exercise search, and plan generation. A single verification command shall run linting, type checking, and the full suite, and shall pass before any change is merged (NFR-MNT-1).

Safety-critical behaviour — the constraint filtering in 5.13.4 and the validation in 5.13.5 — shall additionally be covered by the adversarial suite described in AC-1, which is maintained independently of the unit suite and treated as a release gate.

10.2 Acceptance criteria

ID	Criterion	Method
AC-1	Across a 200-case adversarial suite, zero plans reaching a user contain a contraindicated or allergen-violating item.	Automated suite
AC-2	At least 95% of generations return a schema-valid plan on first attempt.	Staged measurement
AC-3	At least 90% pass the validator without repair.	Staged measurement
AC-4	Quota cannot be exceeded by a modified client.	Direct API testing with tampered client and token
AC-5	Commit never modifies a workout that is not planned.	Transaction test
AC-6	Undo restores the previous schedule exactly.	Automated test
AC-7	95th-percentile latency at or below 12 s under 50 concurrent generations.	Load test
AC-8	The application remains fully functional with the AI service unreachable.	Manual and automated, network disabled
AC-9	No health or condition data appears in any analytics payload.	Network capture inspection
AC-10	A prompt-injection attempt in the free-text note cannot cause an excluded item to be recommended.	Adversarial suite

AC-1 must be built before implementation begins, not after. It is the only mechanism that converts the guarantee described in 5.13.1.4 from an intention into a verified property. A suite written after the fact tends to encode the behaviour that was built rather than the behaviour that was required.

11 Risks

ID	Risk	Severity	Mitigation
R-1	A contraindicated exercise is recommended and a user is injured.	Critical	SQL-level filtering (FR-AIA-6), deterministic validation (FR-AIA-9), AC-1 suite, qualified rule ownership (Q4)
R-2	Store rejection on medical-claim grounds.	High	Section 8 complete before submission; pre-submission legal review
R-3	Catalogue metadata is thin, so filtering under-blocks.	High	Phase 1 content gate; every rejection logged to locate gaps (FR-AIA-12)
R-4	Cost overrun through abuse.	Medium	Server-side quota, rate limits, global ceiling (FR-AIA-24)
R-5	Model provider outage.	Medium	Graceful degradation (NFR-REL-2), quota preserved, secondary provider evaluated
R-6	Prompt regression silently degrades quality.	Medium	Versioned prompts, acceptance suite in CI on every prompt change
R-7	Health data reaches the provider without consent.	High	Allow-list transmission (NFR-PRV-3), FR-CMP-6, AC-9
R-8	Users treat plans as medical advice.	High	Section 8.1 copy discipline, disclaimers, professional-consultation prompts
R-9	Nutrition scope doubles Phase 0 and delays launch.	Medium	Consider training-only first release (Q3)
R-10	GPL obligations conflict with commercial intent for the service tier.	Medium	Resolve FR-CMP-16 with counsel before commercial commitments

R-1 is the risk that governs the architecture of this module. Every design decision in 5.13 — filtering before generation rather than instructing during it, validating after generation, rejecting rather than repairing safety violations, and requiring a named clinical owner — exists to reduce it. The controls are defence in depth precisely because no single one of them is sufficient.

12 Phasing

Phase	Contents	Exit gate
0	Nutrition schema, profile schema, condition vocabulary, onboarding flow	Section 4.2 gaps closed; onboarding shipped and skippable
1	CMS with lifecycle, roles, and audit; content team loads catalogue	At least 500 foods and 300 exercises published with conditional metadata, reviewer-approved
2	AI service: retrieval, generation, validator, quota. No client UI.	AC-1 through AC-4 and AC-10 pass against the service directly
3	Client flow: entry point, intent capture, preview, commit, undo	AC-5, AC-6, AC-8 pass
4	Compliance copy, privacy policy, store disclosures, provider terms, DPIA	Section 8 complete; legal sign-off
5	Staged rollout 5% to 25% to 100%, monitoring cost and rejection rate	AC-7 holds in production; no AC-1 violations observed

Phase 0 is on the critical path and is substantially larger than it appears from the brief. It comprises a complete onboarding flow plus an entire domain — nutrition — that the core product does not model in any form. Estimating this phase from the AI feature description alone will underestimate it considerably.

Phase 2 deliberately precedes any client work. Building the service first, and proving its safety properties against the acceptance suite before a single screen exists, means the safety-critical component is validated in isolation rather than through the interface.

13 Open Decisions

These require resolution before Phase 2. Each materially changes implementation.

#	Question	Consequence
Q1	Is quota three per month universally, or does a paid tier receive more?	Determines whether billing, receipt validation, and subscription management enter v1 scope
Q2	What is the plan horizon — one day, seven days, or twenty-eight?	Drives token budget, cost per generation, and validator complexity
Q3	Does v1 cover nutrition and training together, or training only?	Nutrition roughly doubles Phase 0; training-only is a materially faster path to launch
Q4	Who signs off that a contraindication rule is clinically sound?	Currently unassigned; highest-severity open item in this document
Q5	Which markets at launch?	EU launch triggers GDPR Article 9 obligations and a DPIA (FR-CMP-13)
Q6	Is the CMS a bespoke build or an off-the-shelf headless CMS with a custom schema?	Approximately six weeks of schedule difference
Q7	Is the exercise media bucket and sync provider infrastructure owned and provisioned?	Exercise media hosting and a sync provider must both be provisioned before launch

13.1 A note on Q4

Every control specified in 5.13.5 assumes that the catalogue's contraindication data is correct. The system will enforce those rules faithfully and verifiably. It cannot determine whether the rules themselves are clinically sound.

That judgement requires a qualified human owner with named accountability. The originating brief does not identify one. Until it does, the safety architecture described in this document is structurally complete but resting on an unvalidated foundation, and R-1 is not genuinely mitigated regardless of how well the software behaves.

14 Appendix A — Requirement Summary

Area	Prefix	Count	Section
Architecture	FR-ARC	2	3.2
Identity and accounts	FR-IDN	4	5.1
Profile and onboarding	FR-PRO	9	5.2
Exercise catalogue	FR-EXC	6	5.3
Workout planning	FR-WKP	5	5.4
Workout execution	FR-WKE	8	5.5
Body measurements	FR-MEA	4	5.6
Results and analysis	FR-RES	4	5.7
Health integration	FR-HLT	5	5.8
Watch and Live Activity	FR-WAT	4	5.9
Notifications	FR-NOT	3	5.10
Settings and localisation	FR-SET	6	5.11
Synchronisation	FR-SYN	6	5.12
AI Training Assistant	FR-AIA	33	5.13
Content Management System	FR-CMS	11	5.14
Data	FR-DAT	3	4.3
Interfaces	FR-INT, FR-UI	6	6.1, 6.3
Compliance	FR-CMP	16	8
Non-functional	NFR-*	46	7, 9

Approximate totals: **135 functional requirements** and **46 non-functional requirements**.

The AI Training Assistant accounts for 33 functional requirements directly, and is the originating cause of a further 39 across the profile, data, CMS, and compliance areas. Estimating the module from Section 5.13 alone will therefore understate it by roughly a factor of two.

15 Appendix B — Traceability

Risk	Mitigating requirements	Verifying criteria
R-1 Injury from contraindicated content	FR-AIA-6, FR-AIA-9, FR-AIA-10, FR-AIA-12, FR-CMS-4, FR-CMS-5, FR-CMS-7	AC-1, AC-10
R-2 Store rejection	FR-CMP-1 to FR-CMP-9	Section 8 review
R-3 Thin catalogue metadata	FR-CMS-4, FR-CMS-6, FR-CMS-9, FR-AIA-12	Phase 1 gate
R-4 Cost overrun	FR-AIA-20 to FR-AIA-25	AC-4
R-5 Provider outage	NFR-REL-2, NFR-REL-6, FR-AIA-11	AC-8
R-6 Prompt regression	FR-AIA-3, FR-AIA-4, NFR-MNT-4	AC-2, AC-3
R-7 Health data leakage	NFR-PRV-2, NFR-PRV-3, FR-CMP-6, FR-HLT-5	AC-9
R-8 Plans read as medical advice	FR-CMP-1 to FR-CMP-4, NFR-USA-6	Section 8 review

16 Appendix C — Document Control

Version	Date	Author	Change
1.0	2026-08-07	Engineering	Initial complete specification, covering the shipped application and the AI Training Assistant module

Status: Draft. Sign-off is blocked on the open decisions in Section 13, of which Q4 is the highest severity.